LA-UR-23-26218
Accepted Manuscript

Does Magnetic Reconnection Occur in the Near Lunar Surface Environment?

Sawyer, R. P.
Halekas, J. S.
Bonnell, J. W.
Chen, L. J.
McFadden, J.
Glassmeier, K. H.
Harada, Y.
Stanier, Adam John

Provided by the author(s) and the Los Alamos National Laboratory (2024-09-16).

To be published in: Geophysical Research Letters

DOI to publisher's version: 10.1029/2023GL104733

Permalink to record:
https://permalink.lanl.gov/object/view?what=info:lanl-repo/lareport/LA-UR-23-26218

Los Alamos National Laboratory, an affirmative action/equal opportunity employer, is operated by Triad National Security, LLC for the National Nuclear Security Administration of U.S. Department of Energy under contract 89233218CNA000001. By approving this article, the publisher recognizes that the U.S. Government retains nonexclusive, royalty-free license to publish or reproduce the published form of this contribution, or to allow others to do so, for U.S. Government purposes. Los Alamos National Laboratory requests that the publisher identify this article as work performed under the auspices of the U.S. Department of Energy. Los Alamos National Laboratory strongly supports academic freedom and a researcher's right to publish; as an institution, however, the Laboratory does not endorse the viewpoint of a publication or guarantee its technical correctness.

Check for updates

Geophysical Research Letters

RESEARCH LETTER
10.1029/2023GL104733

Does Magnetic Reconnection Occur in the Near Lunar Surface Environment?

R. P. Sawyer[1], J. S. Halekas[1], J. W. Bonnell[2], L. J. Chen[3], J. McFadden[2], K. H. Glassmeier[4], Y. Harada[5], and A. Stanier[6]

[1]Department of Physics and Astronomy, University of Iowa, Iowa City, IA, USA, [2]Space Sciences Laboratory, University of California, Berkeley, Berkeley, CA, USA, [3]NASA Goddard Space Flight Center, Greenbelt, MD, USA, [4]Technical University of Braunschweig, Braunschweig, Germany, [5]Department of Geophysics, Graduate School of Science, Kyoto University, Kyoto, Japan, [6]Los Alamos National Laboratory, Los Alamos, NM, USA

Key Points:

- Observations suggest magnetic reconnection occurs between the solar wind IMF and lunar crustal magnetic fields
- Electron pitch angle and velocity distributions suggest the spacecraft traversed a closed magnetic topology containing solar wind electrons
- We report in-situ observations of demagnetized ions and associated Hall electric fields near the lunar surface

Correspondence to:
R. P. Sawyer,
rpsawyer@uiowa.edu

Citation:
Sawyer, R. P., Halekas, J. S., Bonnell, J. W., Chen, L. J., McFadden, J., Glassmeier, K. H., et al. (2023). Does magnetic reconnection occur in the near lunar surface environment? *Geophysical Research Letters*, *50*, e2023GL104733. https://doi.org/10.1029/2023GL104733

Received 5 JUN 2023
Accepted 15 AUG 2023

© 2023. The Authors.
This is an open access article under the terms of the Creative Commons Attribution-NonCommercial-NoDerivs License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

Abstract

The near lunar surface contains small-scale magnetic field structures that provide a natural test bed for observing plasmas with a non-zero Hall electric field, as well as potentially facilitating electron-only reconnection. This study presents observational evidence of magnetized electrons as well as demagnetized ions when THEMIS-ARTEMIS probe B reached an altitude of ∼15 km above the lunar surface. Additionally, observations suggest the presence of a field line topology change and traversal of a closed magnetic field structure containing solar wind electrons, suggestive of magnetic reconnection having occurred at some point between the solar wind interplanetary magnetic field and a lunar crustal magnetic field. Thus, the observations presented here are consistent with previous studies that predict prominent Hall electric fields near lunar crustal magnetic fields and further suggest that the solar wind interplanetary magnetic field may reconnect with lunar crustal magnetic fields, most likely via electron-only reconnection.

Plain Language Summary

While interactions between the solar wind and the Earth's magnetosphere have been well studied, there is still much to be learned by studying the interactions between the solar wind and the small-scale lunar magnetic fields. Due to the small-scale nature of the lunar magnetic fields, previous studies have suggested that the ions do not respond in the same manner as the electrons. The resulting effects lead to an electric field near regions of lunar magnetic fields. This study presents observational evidence of the aforementioned phenomena. Additionally, the spacecraft observations also suggest that magnetic reconnection, or the breaking of the lunar magnetic field lines and reconnection to the magnetic field in the solar wind, was occurring between the solar wind and the lunar magnetic fields.

1. Introduction

In typical space plasmas such as in the solar wind the Hall term in the generalized Ohm's law is near zero. This is because the ions and electrons both remain magnetized in a plasma without small-scale gradients. However, near the reconnecting magnetopause within ion diffusion regions the Hall term can be non-zero (Mozer et al., 2002, 2008). Non-zero Hall electric fields also may occur near the lunar surface (Jarvinen et al., 2014; Saito et al., 2012). Near the lunar surface the scale size of the crustal magnetic field is smaller than that of the ion gyroradius and thus the ions are not capable of responding to the small-scale Lorentz forces that are developed, while the electrons remain magnetized and respond accordingly.

Saito et al., 2012 suggested that if the ions were demagnetized near the lunar surface while the electrons were still magnetized, the resulting Hall electric field could explain the simultaneous ion deceleration and electron acceleration observed near crustal magnetic fields. However, the Saito et al., 2012 study does not explicitly show the demagnetized ions, but rather infers the presence of the Hall electric field. Jarvinen et al., 2014 further showed the generation of a Hall electric field by the demagnetization of ions with use of a 3-D quasi-neutral hybrid simulation with the ions as particles and the electrons acting as a charge neutralizing fluid. The results suggest that the Hall electric field is directly associated with lunar crustal magnetic fields. These two studies indicate that the near lunar surface environment exhibits an important characteristic, the ions are demagnetized—yet no direct evidence has been reported up to date.

In addition to the Hall effects, previous studies have also suggested that reconnection may occur between the solar wind IMF and lunar crustal magnetic fields. Simulation and modeling studies (Deca et al., 2015; Giacalone

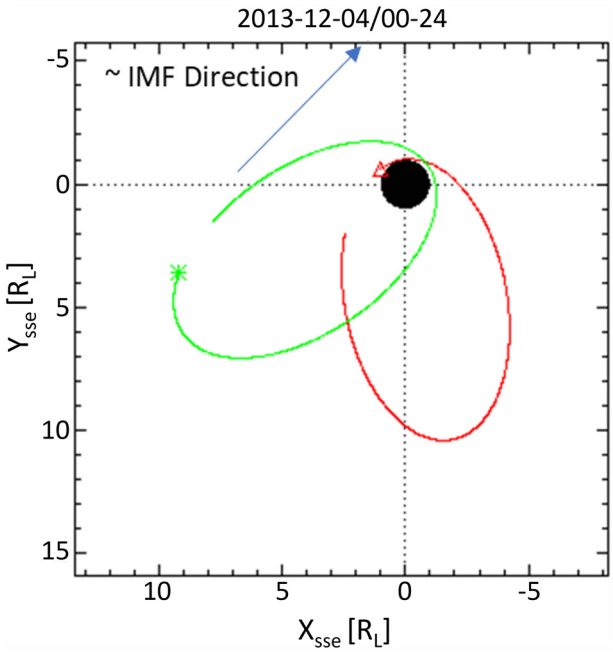


Figure 1. The orbital configuration of THEMIS-B (THB) is shown in red and THEMIS-C (THC) is shown in green for the orbit on 04 December 2013 in the X-Y plane. The approximate IMF direction was determined from THC and is shown as a blue arrow. Coordinate units are in terms of Lunar Radii (R_L).

& Hood, 2015; Harnett & Winglee, 2000, 2003) have explored the interaction between the solar wind IMF and plasma flow with lunar crustal fields. The results showed field line topologies suggestive of reconnection having occurred. Interestingly, kinetic-ion fluid-electron hybrid simulations of Giacalone & Hood, 2015 found that reconnection exhausts like those expected for typical reconnection did not occur in their simulations. Also, electron distributions near a magnetic null region do not exhibit characteristics like those found in electron diffusion regions at the terrestrial magnetopause, that is, agyrotropic electron distributions were not observed (Deca et al., 2015; Harnett & Winglee, 2000, 2003). Typically, agyrotropic electron distributions are associated with reconnection and have been observed within electron diffusion regions. Agyrotropic electron distributions are driven by finite electron gyroradius effects as well as the demagnetization of the electrons within the electron diffusion region. This results in asymmetries in the perpendicular electron temperature and can generate a "crescent" shape in the electron velocity distribution function (Bessho et al., 2016, 2017; Hesse et al., 2014, 2016). Open questions remain as to the nature and whether reconnection occurs between the solar wind IMF and lunar crustal magnetic fields. As a result, the exploration of reconnection with in-situ orbiting spacecraft near the lunar surface is of considerable interest.

The focus of this study is to assess whether reconnection may have occurred between the IMF and lunar crustal magnetic fields, and determine if the environment supports the demagnetization of ions in the near lunar surface environment. In order to accomplish these tasks, we utilized in-situ electron and ion plasma observations as well as electric and magnetic field measurements made by the THEMIS-ARTEMIS spacecraft in the near lunar surface environment, during an orbit when the spacecraft reached an altitude at periselene near 15 km on 04 December 2013.

2. THEMIS-ARTEMIS Instrumentation and Observations

This study presents an event from 04 December 2013 when the THEMIS-ARTEMIS Probe B (THB) made near-lunar surface measurements at altitudes of 10s of km near the terminator. THEMIS-ARTEMIS Probe C (THC) was in the solar wind and acted as a near moon solar wind monitor for the duration of the event. The orbits of the spacecraft are shown in Figure 1 along with the approximate IMF projection on the X-Y plane. The X-Z plane is not shown as THB was in a primarily equatorial orbit.

We utilized data from the ion and electron top-hat electrostatic analyzers (ESA) (McFadden et al., 2008), electric field double probes (EFI) (Bonnell et al., 2008), and the fluxgate magnetometer (FGM) (Auster et al., 2008). Electron burst mode data was available from the ESA, allowing for approximately 4 s time steps and full energy and solid angle coverage. We utilized the DC electric field measurements from the EFI double probes. Only data from the spin-plane probes were used, while data from the axial probe was disregarded due to uncertainties resulting from the shorter probe length and the probe shorting factor (see Bonnell et al., 2008 for details on this error source). Additionally, the spin-plane probe measurements required a constant offset voltage. We applied a 2.5 mV/m offset from 00:08:00. The 2.5 mV/m offset was determined from the typical offsets found in the Bonnell et al., 2008 study. We further increased the offset during the solar wind interval to 4.5 mV/m in order to make the $E \times B$ velocity zero within the solar wind. The FGM data utilized in this study provided the full 3D-vector magnetic field. The data from THC was shifted in time to match the data from THB. This was accomplished by aligning structures in the solar wind that were observed by both spacecraft.

Figure 2 shows the observations from 04 December 2013 near lunar surface approach of THB with a periselene altitude near 15 km. The spacecraft starts in the solar wind, from 00:05:45 to 00:08:00 UT. During this time the magnetometer data from Figures 1b and 1c showed very small differences between the two spacecraft. Furthermore, the ion flux peaked near 1 keV, typical of solar wind protons. The electrons are present from about 10 eV to 1 keV and exhibit a mostly parallel streaming population. Slightly before 00:08:00 UT the IMF becomes

19448007, 2023, 16, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2023GL104733 by Los Alamos National Laboratory, Wiley Online Library on [16/09/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

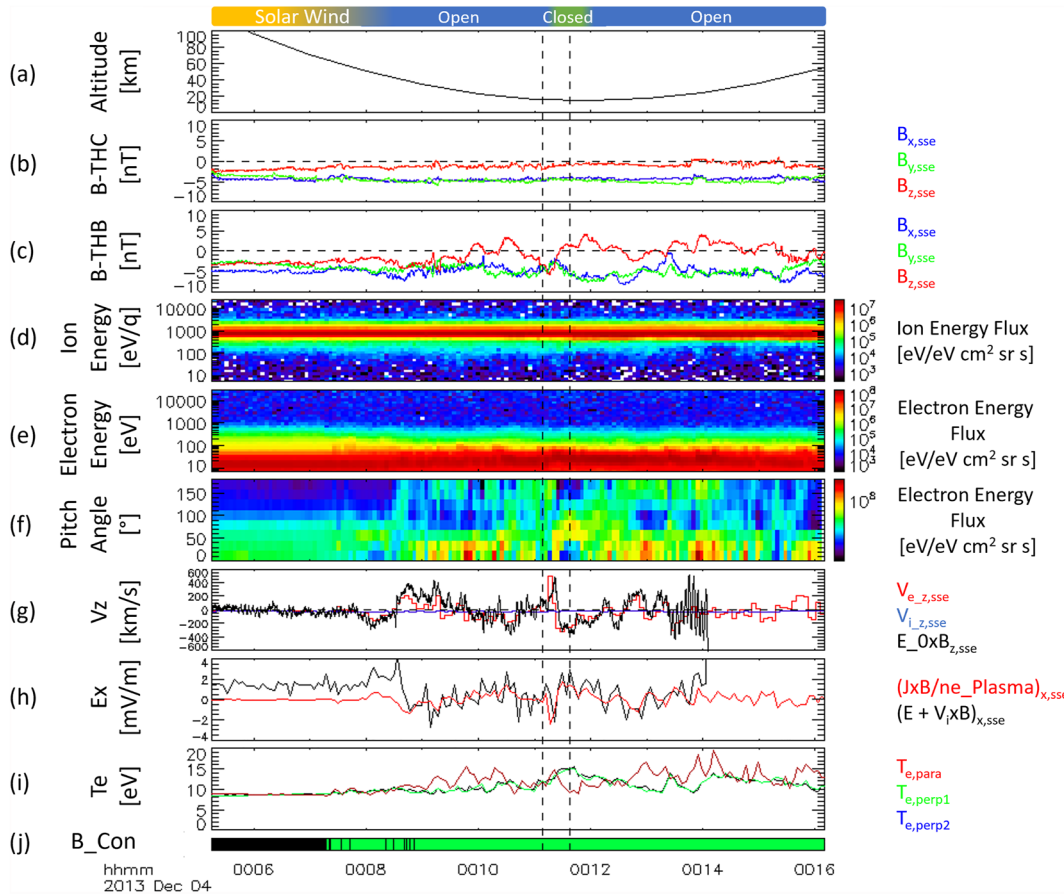


Figure 2. Observations from the 04 December 2013 orbit where the periselene was near 15 km. Panel (a) shows the altitude, panel (b) shows the time-shifted magnetometer data from THC which was used as an upstream solar wind monitor, panel (c) shows the magnetometer data from THB which shows the modification from THC due to crustal field interactions, panel (d) shows the ion energy-time spectrogram highlighting the solar wind ion beam at 1 keV and the reflected/backscattered ions near the lunar surface at energies around 100's of eV, panel (e) shows the electron energy-time spectrogram highlighting the electron energy spectra throughout the periselene, panel (f) shows the pitch angle distribution of electrons within an energy range of 10–300 eV which shows the field line topology change, panel (g) shows the $V_{i,z}$, $V_{e,z}$, and $(E \times B)_z$ components of the respective plasma bulk velocity and field line convection velocity (h) comparison of $(E + V_i \times B)_x$ and the plasma derived Hall electric field $(J \times B/ne)_x$; panel (i) shows the electron temperature profile indicating regions of electron heating, and finally panel (j) shows the magnetic connection to the surface using a straight field line approximation, with green meaning connectivity to the moon. Dashed lines represent times in which velocity distributions were obtained. Open field lines are defined as having a single footpoint in the lunar regolith, whereas a closed field line topology is defined as having both footpoints in the lunar regolith.

connected to the lunar surface and we observe a parallel loss cone. At this time $V_{e,z}$ also starts to become modified relative to $V_{i,z}$. From 00:08:00 to 00:11:10 UT the spacecraft is magnetically connected to the lunar surface and the electron energy distribution exhibits some energization and heating, also apparent in the electron temperature profile. During this time the electron pitch angle distribution takes on a counter-streaming distribution, indicative of an open field line topology. From Figure 1 we can conclude that, as the solar wind flows over the moon, the anti-parallel end of the IMF becomes embedded in the lunar regolith. The parallel moving electrons are thus moving away from the lunar surface and represent lunar sourced and reflected electrons from the lunar surface. The anti-parallel moving electrons are moving toward the lunar surface and represent incident solar wind electrons. During this period the electrons also exhibit heating. During this time $V_{e,z}$ exhibits large variations not seen in $V_{i,z}$, but consistent with $(E \times B)_z$. The magnetic field from THB shows differences from the measurements of THC, indicating a localized region of lunar crustal field influence. From 00:11:10 to 00:12:00 UT the electron pitch angle distribution shifts and is peaked near 90° with a double-sided loss cone distribution at 0° and 180°, indicative of a closed field line topology. During this period of closed field line topology, the electrons are heated.

19448007, 2023, 16, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2023GL104733 by Los Alamos National Laboratory, Wiley Online Library on [16/09/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

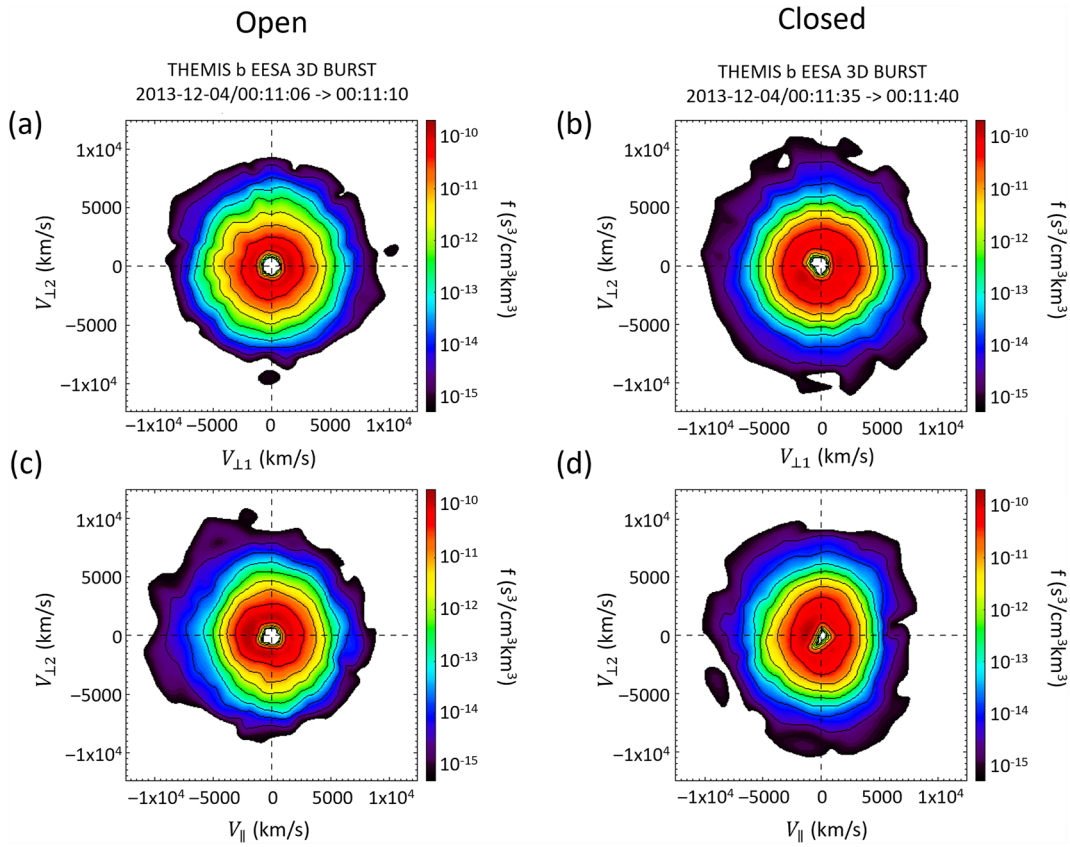


Figure 3. Electron velocity distributions measured at the times corresponding to the dashed lines from Figure 2 during the open-closed field line traversal. Panels (a), (b) show the electron distributions in the perpendicular plane using the $V_{e,bulk}$ subtracted reference frame. The x-axis is in the $V_{\perp 1} = (V_{e,bulk} \times B) \times B$ direction while the y-axis is in the $V_{\perp 2} = -(V_{e,bulk} \times B)$ direction. Panels (c), (d) show the electron distributions along the magnetic field direction using the $V_{e,bulk}$ subtracted reference frame. The x-Axis is in the parallel direction, while the y-axis is in the $V_{\perp 2} = -(V_{e,bulk} \times B)$ direction.

From 00:12:00 to 00:16:00 UT the electron pitch angle distribution exhibits a counter-streaming distribution, indicative of an open field line configuration. THC electron pitch angle distributions throughout this time period (not shown) showed no evidence of being connected to the moon or the double-sided loss cone as observed from THB. As a result, the magnetic field topology description is unique to THB and is directly associated with lunar interactions.

Figure 3 highlights the electron distributions as the spacecraft traversed from open to closed field lines. From Figures 3a and 3b, the electron distributions appear gyrotropic. From Figure 3c there is a slight parallel enhancement which is driven by the counter-streaming nature of the electrons on an open field line topology. Figure 3b also exhibits a mostly gyrotropic electron distribution with a similar energy structure to that of Figure 3a, indicating that solar wind electrons are contained on the closed field line structure. Figure 3d shows a reduction in parallel flux when compared to Figure 3c which is driven by the double-sided loss cone shown in Figure 2.

3. Results and Discussion

From Figures 2 and 3 we draw two results: (a) the closed field line topology contains electrons with a similar energy distribution to that of the solar wind. (b) The electrons remain magnetized, while the ions are demagnetized.

The first result from this event comes from the traversal of the spacecraft onto a closed magnetic field topology. During the traversal of the region marked "closed" in Figure 2 the electron pitch angle distributions indicated a double-sided loss cone distribution. This distribution indicates that both footpoints of the magnetic field line passing through the spacecraft are embedded in the lunar regolith. As a result, the electrons are mirrored near

19448007, 2023, 16, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2023GL104733 by Los Alamos National Laboratory, Wiley Online Library on [16/09/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

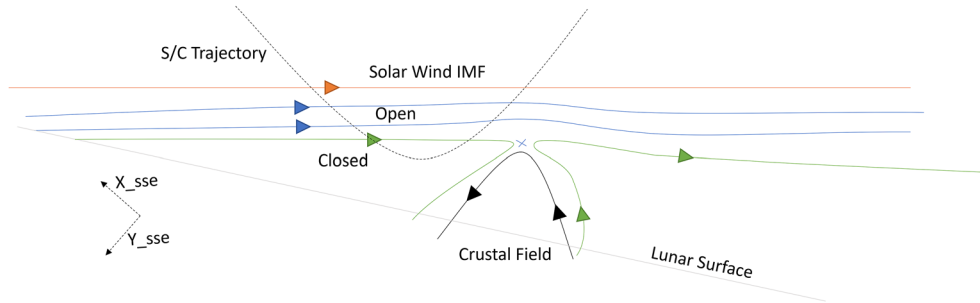


Figure 4. Schematic representing a possible reconnection geometry. The black field lines represent lunar crustal magnetic fields, while the blue field lines represent the IMF. The green field lines represent the IMF field lines reconnected with the crustal fields.

the footpoints and undergo bounce motion. Those electrons with pitch angles more field-aligned than a critical angle are lost to the surface, much as in the Earth's magnetosphere. Figure 3 suggested that solar wind electrons were observed on the closed field line structure. We can speculate as to how solar wind material can access a closed magnetic structure on the lunar surface.

To observe a closed field line structure with solar wind electrons, one possibility is that an open field line structure underwent a topology change to become closed. For this to occur, reconnection must have happened between the solar wind IMF and a lunar crustal magnetic field at some point on the lunar surface. It is unlikely that the observed solar wind electrons could simply gyrate onto the closed field line structure. In an ∼10 nT field, the gyroradius of a 10 eV electron is ∼1 km. The closed field line structure is traversed in ∼20 s leading to an encounter of the closed field line region which was ∼40 km in length, an order of magnitude larger than the electron gyroradius. Additionally, the vertical thickness of the open region prior to the traversal of the closed field line region corresponded to a vertical thickness of ∼35 km, also an order of magnitude larger than the electron gyroradius. Therefore, the solar wind electrons on the closed field line structure are potentially afforded access to the region through reconnection. Figure 4 shows a schematic of a possible reconnection geometry and spacecraft trajectory. An alternate possibility is that secondary electron emission from the surface could give rise to an electron population trapped on a closed field line structure. This scenario would therefore not require magnetic reconnection to take place. However, while secondary electron emission may be occurring it is unlikely that the solar wind electrons would be observed throughout the structure as the electron gyroradius was much less than the observed closed field line structure thickness. Therefore, magnetic reconnection is required to explain the solar wind electron populations present.

On further inspection of the structure of the velocity distributions in Figure 3 and the perpendicular electron temperature profile in Figure 2i, the electron distributions in the perpendicular plane appear mostly gyrotropic. There are no obvious crescent shaped features indicative of an agyrotropic electron distribution within an electron diffusion region. Additionally, the similarity in the perpendicular electron temperature profiles, as shown in Figure 2i, further support the gyrotropic nature of the electron distributions. Since there is no significant agyrotropy, as expected near the separatrix or the electron diffusion region., it is likely that either a) the electron diffusion region is not near the spacecraft location and so we do not observe any finite gyroradius effects or b) more likely the temporal and therefore spatial resolution of the THEMIS-ARTEMIS spacecraft is inadequate to resolve the finite gyroradius effects that drive the agyrotropic electron distributions.

The second result comes from the $V_{i,z}$ and $V_{e,z}$ measurements. From Figure 2g, $V_{i,z}$ and $V_{e,z}$ are not equal outside of the solar wind interval. Some disagreement between $V_{i,z}$ and $V_{e,z}$ could result from errors related to the difficulty in determining the bulk electron velocity from the electron distributions. However, comparison of $V_{i,z}$ and $V_{e,z}$ with the $(E \times B)_z$ velocity suggests a different interpretation. There is agreement between $V_{e,z}$ and $(E \times B)_z$, supporting the result that the electrons are magnetized. On the other hand, the lack of a response in $V_{i,z}$ indicates that the ions are demagnetized within this region. One of the key results of the Saito et al. (2012) and Jarvinen et al. (2014) studies depended on the ions becoming demagnetized near the lunar surface, thus generating the Hall electric field proposed by both studies. Our second result provides experimental evidence of the proposed Hall electric field in these studies. The results presented here showcase a very similar environment to that found near the magnetopause by Mozer et al., 2008. That study found that there was a region near the reconnecting

19448007, 2023, 16, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2023GL104733 by Los Alamos National Laboratory, Wiley Online Library on [16/09/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

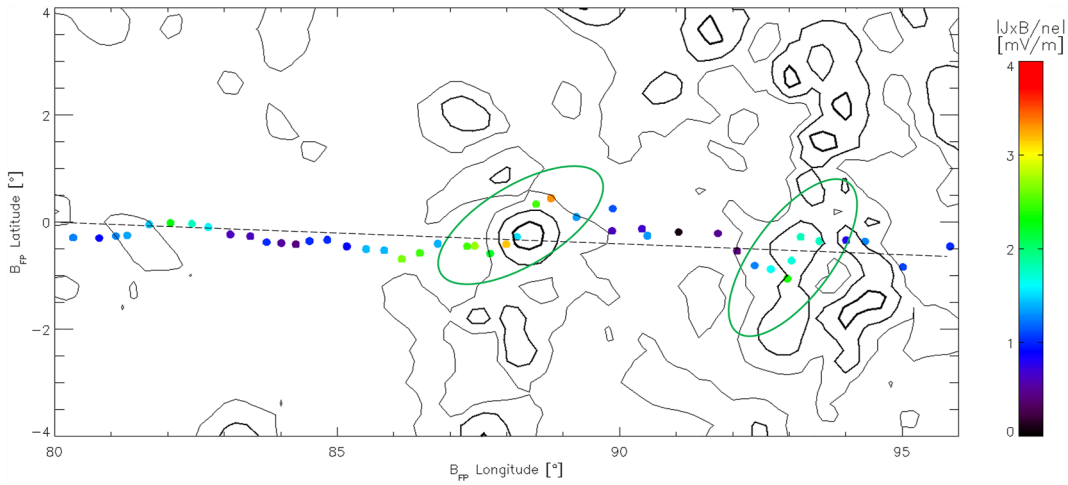


Figure 5. The magnitude of the plasma derived Hall electric field generated by the decoupling of ion and electron motion, as computed from the bulk ion and electron velocities. The crustal field map is given by the Tsunakawa et al., 2015 crustal field model at the surface and drawn at 6, 10, and 15 nT contours. The green contours highlight where the Hall electric field is enhanced near regions of crustal magnetic field strength.

magnetopause where the ion bulk velocities perpendicular to the magnetic field disagreed while the electron perpendicular velocity agrees with the $E \times B$ velocity, similarly indicating a region where Hall MHD is applicable.

Mozer et al. (2008) showed that the electric field measured by the EFI instrument agreed with the Hall electric field. This study presents a very similar result near the lunar surface. When $V_{e,z}$ is consistent with $(E \times B)_z$, we can utilize the bulk ion and electron velocities to calculate the Hall electric field from the plasma data (i.e., $J = V_i - V_e$). Figure 2h shows a correspondence between the $E + V_i \times B$ and the Hall electric field, consistent with the differential motion between the ions and the electrons generating a Hall electric field within this region. The magnitude of the calculated Hall electric field is shown in Figure 5 over a map of the local crustal field magnitude. Figure 5 shows that the Hall electric field magnitude is enhanced near regions of relatively higher local crustal field strength. This result qualitatively agrees with previous predictions of Hall electric fields driven by differential motion between ions and electrons at crustal fields (Jarvinen et al., 2014; Saito et al., 2012). However, from Figure 2d there is no discernible ion deceleration and Figure 2i suggests electron heating as opposed to acceleration. The lack of ion deceleration could be due to the fact that this event was over a particularly weak crustal magnetic field compared to the event described in Saito et al., 2012.

From the results presented in this study it is most likely that the reconnection process at play would be electron-only reconnection (Phan et al., 2018; Stawarz et al., 2019, 2022). This assertion is driven by the fact that the small scale size of the crustal field regions may provide a natural regime for electron-scale current sheets to form that are not embedded in a larger ion-scale current sheet. During the electron-only reconnection events exhibited in previous studies, no modification of the ion dynamics were reported. Thus, the lack of ion acceleration, as expected in traditional reconnection, would be accounted for in this scenario. As a result, the region in the near lunar surface environment may naturally cause the resulting plasma regime to be in a state where the ions do not participate in the reconnection process.

4. Conclusions

In conclusion, the THEMIS-ARTEMIS observations described in this study reveal the following characteristics:

1. The closed field line topology containing solar wind electrons may indicate the occurrence of reconnection at some point between the solar wind IMF and a lunar crustal magnetic field.
2. The region near the lunar surface contains demagnetized ions and magnetized electrons that are $E \times B$ drifting, resulting in a Hall electric field associated with regions of relatively stronger crustal magnetic field strength.
3. Given the demagnetized nature of the ions and the relevant spatial scale being a small fraction of an ion inertial length, it is most likely that the type of reconnection that occurs in the near lunar surface environment is electron-only reconnection.

19448007, 2023, 16, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2023GL104733 by Los Alamos National Laboratory, Wiley Online Library on [16/09/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Given the above conclusions we can state that the near lunar surface environment can support the demagnetization of ions as previously predicted. From the observed closed magnetic field topology containing solar wind electrons we posit that magnetic reconnection may have occurred at some point between the IMF and a lunar crustal magnetic field. Future work and low-altitude measurements at a higher temporal resolution are needed to explore further the nature of the reconnection process in this environment. For instance, what are the factors of import that can suppress (Doss et al., 2015, 2016; Swisdak et al., 2003) magnetic reconnection in the plasma regime found near lunar crustal magnetic fields? Given the new interest in electron-only reconnection, it is of great interest to the community to utilize the Moon as a natural laboratory to explore the fundamental nature of reconnection.

Data Availability Statement

All ARTEMIS data used in this paper are publicly available at: http://artemis.ssl.berkeley.edu/data_files.shtml. Data processing was handled through IDL routines that are publicly available in the current SPEDAS software package, which can be found through the SPEDAS website (http://spedas.org).

Acknowledgments
Research at the University of Iowa was funded by NASA through NASA contract NAS5-02099 and the Solar System Exploration Research Virtual Institute (80NSSC20M0022). We acknowledge NASA contract NAS5-02099 and V. Angelopoulos for use of data from the THEMIS Mission. K. H. Glassmeier is financially supported by the German Ministerium für Wirtschaft und Klimaschutz and the Deutsches Zentrum für Luft und Raum-fahrt under Contract 50 OC 1403.

References

Auster, H. U., Glassmeier, K. H., Magnes, W., Aydogar, O., Baumjohann, W., Constantinescu, D., et al. (2008). The THEMIS fluxgate magnetometer. *Space Science Reviews*, *141*(1), 235–264. https://doi.org/10.1007/s11214-008-9365-9

Bessho, N., Chen, L.-J., & Hesse, M. (2016). Electron distribution functions in the diffusion region of asymmetric magnetic reconnection. *Geophysical Research Letters*, *43*(5), 1828–1836. https://doi.org/10.1002/2016GL067886

Bessho, N., Chen, L.-J., Hesse, M., & Wang, S. (2017). The effect of reconnection electric field on crescent and U-shaped distribution functions in asymmetric reconnection with no guide field. *Physics of Plasmas*, *24*(7), 072903. https://doi.org/10.1063/1.4989737

Bonnell, J. W., Mozer, F. S., Delory, G. T., Hull, A. J., Ergun, R. E., Cully, C. M., et al. (2008). The electric field instrument (EFI) for THEMIS. *Space Science Reviews*, *141*(1), 303–341. https://doi.org/10.1007/s11214-008-9469-2

Deca, J., Divin, A., Lembège, B., Horányi, M., Markidis, S., & Lapenta, G. (2015). General mechanism and dynamics of the solar wind interaction with lunar magnetic anomalies from 3-D particle-in-cell simulations. *Journal of Geophysical Research: Space Physics*, *120*(8), 6443–6463. https://doi.org/10.1002/2015JA021070

Doss, C. E., Cassak, P. A., & Swisdak, M. (2016). Particle-in-cell simulation study of the scaling of asymmetric magnetic reconnection with in-plane flow shear. *Physics of Plasmas*, *23*(8), 082107. https://doi.org/10.1063/1.4960324

Doss, C. E., Komar, C. M., Cassak, P. A., Wilder, F. D., Eriksson, S., & Drake, J. F. (2015). Asymmetric magnetic reconnection with a flow shear and applications to the magnetopause. *Journal of Geophysical Research: Space Physics*, *120*(9), 7748–7763. https://doi.org/10.1002/2015JA021489

Giacalone, J., & Hood, L. L. (2015). Hybrid simulation of the interaction of solar wind protons with a concentrated lunar magnetic anomaly. *Journal of Geophysical Research: Space Physics*, *120*(6), 4081–4094. https://doi.org/10.1002/2014JA020938

Harnett, E. M., & Winglee, R. (2000). Two-dimensional MHD simulation of the solar wind interaction with magnetic field anomalies on the surface of the Moon. *Journal of Geophysical Research*, *105*(A11), 24997–25007. https://doi.org/10.1029/2000JA000074

Harnett, E. M., & Winglee, R. M. (2003). 2.5-D fluid simulations of the solar wind interacting with multiple dipoles on the surface of the Moon. *Journal of Geophysical Research*, *108*(A2), 1088. https://doi.org/10.1029/2002JA009617

Hesse, M., Aunai, N., Sibeck, D., & Birn, J. (2014). On the electron diffusion region in planar, asymmetric, systems. *Geophysical Research Letters*, *41*(24), 8673–8680. https://doi.org/10.1002/2014GL061586

Hesse, M., Liu, Y.-H., Chen, L.-J., Bessho, N., Kuznetsova, M., Birn, J., & Burch, J. L. (2016). On the electron diffusion region in asymmetric reconnection with a guide magnetic field. *Geophysical Research Letters*, *43*(6), 2359–2364. https://doi.org/10.1002/2016GL068373

Jarvinen, R., Alho, M., Kallio, E., Wurz, P., Barabash, S., & Futaana, Y. (2014). On vertical electric fields at lunar magnetic anomalies. *Geophysical Research Letters*, *41*(7), 2243–2249. https://doi.org/10.1002/2014GL059788

McFadden, J. P., Carlson, C. W., Larson, D., Ludlam, M., Abiad, R., Elliott, B., et al. (2008). The THEMIS ESA plasma instrument and in-flight calibration. *Space Science Reviews*, *141*(1), 277–302. https://doi.org/10.1007/s11214-008-9440-2

Mozer, F. S., Angelopoulos, V., Bonnell, J., Glassmeier, K. H., & McFadden, J. P. (2008). THEMIS observations of modified Hall fields in asymmetric magnetic field reconnection. *Geophysical Research Letters*, *35*(17), L17S04. https://doi.org/10.1029/2007GL033033

Mozer, F. S., Bale, S. D., & Phan, T. D. (2002). Evidence of diffusion regions at a subsolar magnetopause crossing. *Physical Review Letters*, *89*(1), 015002. https://doi.org/10.1103/PhysRevLett.89.015002

Phan, T. D., Eastwood, J. P., Shay, M. A., Drake, J. F., Sonnerup, B. U. Ö., Fujimoto, M., et al. (2018). Electron magnetic reconnection without ion coupling in Earth's turbulent magnetosheath. *Nature*, *557*(7704), 202–206. https://doi.org/10.1038/s41586-018-0091-5

Saito, Y., Nishino, M. N., Fujimoto, M., Yamamoto, T., Yokota, S., Tsunakawa, H., et al. (2012). Simultaneous observation of the electron acceleration and ion deceleration over lunar magnetic anomalies. *Earth Planets and Space*, *64*(2), 4–92. https://doi.org/10.5047/eps.2011.07.011

Stawarz, J. E., Eastwood, J. P., Phan, T. D., Gingell, I. L., Pyakurel, P. S., Shay, M. A., et al. (2022). Turbulence-driven magnetic reconnection and the magnetic correlation length: Observations from Magnetospheric Multiscale in Earth's magnetosheath. *Physics of Plasmas*, *29*(1), 012302. https://doi.org/10.1063/5.0071106

Stawarz, J. E., Eastwood, J. P., Phan, T. D., Gingell, I. L., Shay, M. A., Burch, J. L., et al. (2019). Properties of the turbulence associated with electron-only magnetic reconnection in Earth's magnetosheath. *The Astrophysical Journal*, *877*(2), L37. https://doi.org/10.3847/2041-8213/ab21c8

Swisdak, M., Rogers, B. N., Drake, J. F., & Shay, M. A. (2003). Diamagnetic suppression of component magnetic reconnection at the magnetopause. *Journal of Geophysical Research*, *108*(A5), 1218. https://doi.org/10.1029/2002JA009726

Tsunakawa, H., Takahashi, F., Shimizu, H., Shibuya, H., & Matsushima, M. (2015). Surface vector mapping of magnetic anomalies over the Moon using Kaguya and Lunar Prospector observations. *Journal of Geophysical Research: Planets*, *120*(6), 1160–1185. https://doi.org/10.1002/2014JE004785

19448007, 2023, 16, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2023GL104733 by Los Alamos National Laboratory, Wiley Online Library on [16/09/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License